



Database System Principles

《数据库系统原理》

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- **Part 0: Overview**
 - Ch1: Introduction
- **Part 1 Relational Languages**
 - Ch2: Relational model
 - Ch3: Introduction to SQL
 - Ch4: Intermediate SQL
 - Ch5: Advanced SQL
- **Part 2 Database Design**
 - Ch6: Database design via E-R model
 - Ch7: Relational database design
- **Part 3 Application Design & Development**
 - Ch8: Complex data types
 - Ch9: Application development
- **Part 4 Big Data Analytics**
 - Ch10: Big data
 - Ch11: Data analytics
- **Part 5 Storage Management & Indexing**
 - Ch12: Physical storage systems
 - Ch13: Data storage structures
 - Ch14: Indexing
- **Part 6 Query Processing & Optimization**
 - Ch15: Query processing
 - Ch16: Query optimization
- **Part 7 Transaction Management**
 - **Ch17: Transactions**
 - Ch18: Concurrency control
 - Ch19: Recovery system
- **Part 8 Parallel & Distributed Database**
 - Ch20: Database system architecture
 - Ch21-23: Parallel & distributed storage, query processing & transaction processing
- **Advanced topics**
 - DB Platform: **OceanBase**, MongoDB, Neo4J
 - RAG, Multimodal retrieval, ...

- **Drawbacks of file-processing system**
 - Data redundancy and inconsistency
 - Difficulty in accessing data
 - Data isolation
 - Integrity problems
 - Atomicity of updates
 - Concurrent access by multiple users
 - Security problems
- Database, Database Management System (DBMS), Database system
- **Abstraction for database**
 - Physical level, logical level, view level
- Schema vs. instance
- DB conceptual/logical design
- **DB users**
 - Different user types
 - The duties of DBA
- Major DMBS systems

- **Concepts**
 - Relational schema, relation, relation instance, tuple, attribute
 - Superkey, candidate key, primary key, foreign key (referencing relation, referenced relation)
 - Query language: procedural, non-procedural
- **Relation algebra**
 - **Six basic operators:** select, project, rename, union, set difference, cartesian product
 - Set intersection, natural join, assignment
 - Generalized projection, outer join, aggregate functions
 - Null values
 - Deletion, insertion, updating
 - Views

- **Components of SQL**
 - DDL & DML
- **Domain types in SQL**
- **Create table construct**
 - Integrity constraints
 - Insertion and deletion
 - Drop and alter
- select, rename, ordering of tuples, duplicates, set operations, aggregation
- Null values
- Nested subqueries
- Set comparison
- Views
- **Modification of the Database**
 - Deletion
 - Insertion
 - Updates

- **Integrity constraints**
 - Domain constrains, unique constrains, referential integrity, check, **assertion**
- **Security and authorization**
 - Levels of security (DB, OS and Network)
 - Forms of authorization to modify DB
 - Granting privileges, role
 - Audit trails, encryption, authentication
- Embedded SQL
 - Update through cursors
- Dynamic SQL
- Functions and procedural constructs
- Recursive queries
- Advanced SQL features

- **Database Design**
 - Conceptual design
 - Logical design
 - Physical design
- **Relationship set**
 - Degree of mapping cardinality
- **ER design issues**
 - use of entity sets vs. attributes
 - use of entity sets vs. relationship sets
 - binary vs. n-ary relationship sets
 - use of specialization/generalization
- Symbols used in E-R diagram
- Reduction of an E-R diagram to relational tables

- **Functional Dependency/函数依赖**

- Closure of functional dependency
 - **Armstrong's axioms**: reflexivity(自反), augmentation(增广), transitivity(传递)
 - **Additional rules**: union(合并), decomposition(分解), pseudotransitivity(伪传递)
- Closure of attribute set
- Canonical cover
 - Extraneous attribute and its testing
- Find all candidate keys

- **Decomposition**

- Lossless-join decomposition & dependency preservation

- **Normalization & Normal Forms (范式)**

- Normalization: 1NF, 2NF, 3NF, BCNF
- Testing for BCNF and 3NF, and the corresponding decomposition algorithms

- **Classification of physical storage media**
 - Speed, cost, reliability, volatile, non-volatile
 - Cache, main memory, flash, magnetic disk, optical storage, tape storage
- **Performance measure and optimization of disk access**
 - Access time, data transfer rate, mean time to failure
 - Block, disk-arm-scheduling
- Storage access and buffer manage
- File organization
 - Fixed-length, variable-length records, organization of records in files
- Data dictionary storage
- Raid levels 1-6



- **Basic concepts**
 - Search key and index file, ordered index and hashed index
 - Equal query and range query
 - Primary index or clustering index, secondary index or non-clustering index
 - Dense, sparse, multilevel indices
- **Ordered index**
 - **B⁺-tree index, B-tree index**
- **Hash index**
 - Static hashing
 - Dynamic hashing: extendable hashing
- **Ordered indexing vs. hashing**
- **Index definition in SQL**
- Multiple-key access: grid files and bitmap index

- **Basic steps in query processing**
 - Parsing and translation, optimization, evaluation
- **Measures of query cost**
 - Seek time, number of data block transfers
- **Selection operation**
 - File scan vs. index scan
 - A1-A10
- **Sorting**
 - External sort-merge

- **Join operation**
 - Nested-loop join
 - Block nested-loop join
 - Indexed nested-loop join
 - Merge-join
 - Hash-join
- **Other operations**
 - Duplicate elimination, aggregation, set operations, outer join
- **Evaluation of expressions**
 - Materialization(物化)
 - Pipelining(流水线)

- **Evaluating a given query**
 - Equivalent expressions: equivalence results
 - Different algorithms: measure of cost
- **Measures of cost**
 - Select size
 - Join size
 - Distinct values
- Execution plan
- Practical query optimizers combine the following two approaches
 - Search all the plans and choose the best plan in a cost-based fashion: dynamic programming
 - Use heuristics to choose a good plan

- **Transaction**
 - What's transaction and the two main issues to deal with?
 - Properties: **ACID**
 - Transaction states
- **Serializability**
 - Conflict serializability
 - View serializability
- Recoverability, cascading rollback, cascadeless schedule
- **Testing for Serializability**
 - Precedence graph

- **Lock-based protocols**
 - Lock-compatibility matrix
 - Deadlock and starvation
 - Two-phase locking protocol
 - Strict two-phase locking
 - Rigorous two-phase locking
 - Lock manager and lock table
 - Graph based protocol: impose a partial ordering
 - Timestamp-based protocols
- **Multiple granularity**
 - Intention lock modes
 - Compatibility matrix with intention lock modes
- **Deadlock handling**
 - Deadlock prevention protocols
 - graph-based protocol
 - timestamp-based protocols
 - schemes use transaction timestamps: wait-die scheme (非抢占) and wound-wait (抢占)
 - timeout-based schemes
 - Deadlock detection
 - wait-for graph

- **Two approaches for recovery**
 - Shadow-paging (影子页)
 - log-based recovery
 - Deferred database modification
 - Immediate database modification
 - Checkpoint
- **Recovery with concurrent transactions**